# INTERNSHIP REPORT

**ON**

# DEVELOPMENT OF AN AI-POWERED ACCOUNTING CHATBOT FOR LEDGER ANALYSIS

Submitted to

**School of Computer Science and Engineering**
**IILM University, Greater Noida, U.P.**

In partial fulfilment of the requirement for the degree of
Bachelor of Technology in Computer Science and Engineering

**Submitted by**
**KESHAV BANGIA**
**Roll Number: CS-2341306**
**Batch: 2023-27**

**Academic Session 2026-27**

## CANDIDATE'S DECLARATION

I, Keshav Bangia, do hereby declare that the internship report titled "Development of an AI-Powered Accounting Chatbot for Ledger Analysis" has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All references have been properly acknowledged, and this report is my own work. It has not been submitted to any other institute for any degree or diploma requirement. I shall be solely responsible for any copyright violation arising from this report.

Signature: *Keshav Bangia*

Student Name: Keshav Bangia

Roll No.: CS-2341306

## ACKNOWLEDGEMENT

I would like to express my sincere gratitude to IILM University, Greater Noida, and the School of Computer Science and Engineering for providing the academic guidance and resources required to complete this project. I am thankful to my faculty mentor for constructive feedback and encouragement during the planning, implementation, and documentation of the work.

I also acknowledge my parents, peers, and everyone who supported me throughout the internship period. Their guidance, motivation, and feedback helped me complete the Accounting Chatbot project and this report.

Signature: *Keshav Bangia*

Student Name: Keshav Bangia

Roll No.: CS-2341306

NIXI

**Issued on:** 3rd August 2026

**CERTIFICATE OF INTERNSHIP COMPLETION**

This certifies that **Keshav Bangia**, from **IILM University Greater Noida** successfully completed an internship at **Nixi Ads**.

**Internship Profile: Data Analyst**

**Internship Duration**: 1st June 2026 to 31st July 2026

During his tenure with Nixi Ads, he supported in deep market research and consistently demonstrated strong problem-solving abilities, excellent teamwork, and productive initiatives.

We commend Keshav's performance and wish him continued success.

**For and on behalf of Nixi Ads**

Nikunj Nagpal

**Authorized Signatory**

**Nikunj Nagpal**
**Founder, Nixi Ads**

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# 4. PROJECT DESCRIPTION

## 4.1 Introduction

The Accounting Chatbot is a data-driven web application that allows users to ask questions about business ledger data in plain English. Instead of requiring users to write SQL queries or manually filter spreadsheets, the system interprets a natural-language question, produces a read-only SQLite query, executes it against the accounting database, and returns a concise business-friendly answer.

The project combines data engineering, database querying, large language model integration, API development, and a browser-based chat interface. It is designed for common accounting enquiries such as GST liability, indirect expenses, customer balances, vendor obligations, branch performance, and high-value ledger activity.

## 4.2 Organization Profile

The internship was completed at Nixi Ads, as an Intern in the Technology Department from 01 June 2026 to 31 July 2026. The organisation issued an internship completion certificate confirming the successful completion of the internship. The Accounting Chatbot was developed in the context of technology-enabled business analysis, where users need timely and reliable access to accounting information.

The solution is designed for an organisation that maintains day-wise ledger transactions across branches and needs a simpler way for finance users and decision-makers to retrieve operational accounting insights. It demonstrates how a conversational application can reduce the technical barrier between accounting data and business users.

## 4.3 Problem Statement

Ledger data is often stored in large Excel files or databases containing many branches, ledger names, account categories, debit values, credit values, and transaction dates. Extracting an answer from this data normally requires accounting knowledge, spreadsheet filtering skills, or SQL expertise. This increases turnaround time and can lead to inconsistent interpretation of debit-credit balances or financial-year dates.

The problem addressed by this project is to provide a safe, convenient, and understandable interface through which a user can ask accounting questions in natural language while the system performs controlled, read-only analysis of the underlying ledger data.

### 4.4 Project Objectives

1. Convert raw day-wise ledger transactions from Excel into a structured SQLite database.
2. Allow users to ask accounting questions in plain English through a web chat interface.
3. Generate SQLite SELECT queries using an LLM guided by a detailed database schema and accounting rules.
4. Prevent data modification by validating that generated queries are read-only.
5. Return short, useful answers with Indian financial-number context where appropriate.
6. Support disambiguation of names that may occur in multiple data columns.

### 4.5 Scope of the Project

The current scope covers analysis of a single SQLite table named ledger_transactions. It supports questions related to branches, states, account categories, subcategories, account groups, account types, ledger names, dates, debit amounts, credit amounts, GST, expenses, customers, vendors, and balances. The system is intended for query and analysis; it does not create vouchers, post transactions, change accounting records, or replace a full enterprise resource-planning system.

The database-preparation process standardizes branch names, removes surrounding whitespace from text fields, normalizes dates, replaces missing debit and credit amounts with zero, and flags suspicious original values greater than five crore before they are zeroed. The prompt also enforces a valid date-range guard for queries that use document_date.

### 4.6 Technologies and Tools Used

| Technology / Tool | Role in the Project |
|---|---|
| Python | Primary programming language for data preparation, API logic, and database access. |
| Pandas | Reads Excel data, cleans columns and values, and prepares records for storage. |
| SQLite | Stores the cleaned ledger_transactions table and runs read-only analytical queries. |
| FastAPI | Provides the REST API endpoint that receives questions and returns answers. |
| OpenAI Python SDK with NVIDIA API endpoint | Connects the application to the Nemotron language model for SQL generation and answer phrasing. |
| HTML, CSS, and JavaScript | Implements the responsive browser chat interface. |
| JSON | Stores the value lookup map used for candidate-match disambiguation. |

***Table 4.1: Technologies and tools used in the Accounting Chatbot***

### 4.7 System Architecture

The application follows a layered request-processing design. The interface collects a user question and sends it to the FastAPI backend. The backend enriches a system prompt with candidate matches from value_lookup.json, requests an SQL query from the language model, validates that the query is a SELECT statement, executes it on the SQLite database, and sends the result to the language model for concise answer generation. The API then returns the question, SQL, raw results, and final answer to the browser.

| Stage | Component | Function |
|---|---|---|
| 1 | Web interface | Captures a plain-English accounting question. |
| 2 | Value lookup | Finds possible matches for branch, state, ledger, and accounting labels. |

| Stage | Component | Function |
|---|---|---|
| 3 | LLM SQL generator | Converts the question into one SQLite SELECT query. |
| 4 | Safety validator | Rejects non-SELECT or potentially modifying SQL. |
| 5 | SQLite executor | Runs the approved query on ledger_transactions and returns rows. |
| 6 | LLM answer generator | Converts query results into a concise accounting response. |
| 7 | Web interface | Displays the answer and optionally the generated SQL. |

***Figure 4.1: Logical request flow of the Accounting Chatbot***

## 4.8 Methodology

1. Study the Excel ledger structure and identify columns needed for accounting analysis.

2. Rename columns to clean snake_case names and clean categorical values, dates, and numerical fields using Pandas.

3. Load the cleaned data into SQLite as the ledger_transactions table and inspect the schema and sample records.

4. Build a JSON lookup of distinct categorical values to help the system distinguish similar branch, state, and ledger names.

5. Write a system prompt that defines the schema, financial-year rule, debit-credit interpretation, date-quality guard, and SQL-only output rule.

6. Develop a FastAPI endpoint that generates SQL, validates it, executes it, and produces a natural-language answer.

7. Create a browser chat interface with example prompts, status feedback, error handling, and an optional generated-SQL view.

8. Test representative questions such as total GST liability, top customers by debit amount, indirect expenses for a branch, and branches with high expenses.

## 4.9 Expected Outcomes

1. Faster access to ledger insights for users who do not know SQL.

2. More consistent handling of accounting balance direction and Indian financial-year dates.
3. A safer analysis workflow because only SELECT queries are allowed to reach the database.
4. Improved usability through a clean conversational interface and suggested questions.
5. A reusable foundation that can later be extended with authentication, user roles, audit logs, dashboard charts, and broader accounting datasets.
6. A production deployment should keep API keys outside source code, use environment variables or a secret manager, restrict CORS to trusted origins, apply stronger SQL parsing and access controls, and record audit logs. These are recommended next steps for hardening the prototype.

### 4.10 Certificates of Completion and Communication Proof

The internship completion certificate issued by Nixi Ads is attached earlier in this report.

## 5. BIBLIOGRAPHY / REFERENCES

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Project source files: main.py, load_data.py, build_lookup.py, query_db.py, index.html, system_prompt.txt, and accounting.db (Accounting Chatbot project).